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PI Name:	Rhim, Won-Kyu Ph.D.		
Project Title:	Measurements of Thermophysical Properties of Molten Silicon and Germanium		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
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COI Name (Institution):	Johnson, William L. (California Institute of Technology)		
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This ground-based research project conducted the following ground-based experiments: (i) Investigation of the effect of low-pressure hydrogen gas to reduce surface oxides and to produce deeply undercooled states of molten germanium. If successful, measurements of all the thermophysical properties of undercooled germanium will be attempted. The properties that will be measured include Density, Thermal expansion coefficient, Surface tension, Viscosity, Specific Heat/Hemispherical total emissivity, Electric conductivity, Thermal Conductivity.
(ii) To measure thermophysical properties of Si-Ge alloys of varied composition. Thermophysical properties measured will basically include most of the properties listed in (i). All these properties were measured as a function of true temperatures. The high temperature electrostatic levitator (HTESL) at JPL/Caltech, and, if necessary, a similar levitation system available at the Marshall Space Flight Center (MSFC) will be used for this work.

Task Description:	(iii) To measure surface tension and viscosity of molten silicon, germanium, and various SixGe1-x alloys will be conducted under conditions where the sample rotation is strictly controlled. The objective of this ground-based program was to measure the thermophysical properties of molten/undercooled silicon, germanium, and Si-Ge alloys using the high temperature electrostatic levitator (HTESL) and clearly assess the needs of the microgravity environment to achieve the objective with higher degrees of accuracy. Over the past ten years, there have been several organized efforts both in the United States and Japan to accurately measure thermophysical properties of molten silicon using various techniques. However, the reported thermophysical properties have failed to show a converging trend. The gravity field could have directly or indirectly affected the property measurements. Molten silicon and germanium are highly reactive with most crucibles. When certain containers were used in a measurement process, the measured properties of the melt showed strong dependence on the container materials. If chemical reactions took place during property measurements, one could not have avoided the effects of contamination to the experimental results. Containerless approach alleviates sample contamination problem by isolating melts from containers. However, levitation experiments conducted in the strong gravitational field suffer from other kinds of problems, which are physical in nature. Sample levitation on the ground base is achieved by applying a strong levitation force on a sample against the force of gravity. Strong levitation forces tend to generate internal flows as well as vibrational and rotational instabilities on a drop. These undesirable instabilities can contribute to increased uncertainties in measurement results. Molten silicon and germanium have viscosity in the range of 1mPa·s. Transport properties of such low viscosity liquids can be readily affected by the external perturbation forces. These problems, however, may be greatly reduced by conducting experiments in the low-g environment where the perturbation forces originating from the sample levitation can be reduced by several orders of magnitude.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Study of nucleation/solidification processes and accurate measurements of various thermophysical properties of high temperature liquids are two of the most important NASA exploration goals that were outlined in the NRA documents. In addition, studying the feasibility of accurate measurements of various thermophysical properties of molten/undercooled semiconductors was another important NASA goal. It is safe to say that the HTESL that has been developed by JPL/Caltech under NASA support is the best containerless materials processing facility for high temperature materials research that has already been well-demonstrated by various ground laboratories. It is also expected to operate well in the microgravity environment making those experiments possible that may be impossible under the earth's gravitational influence. Silicon and germanium are the two most important semiconductors for industrial applications: silicon is unsurpassed as a microelectronics material, occupying more than 95% of the electronics market. Si-Ge alloy is attracting keen interest for advanced electronic and optoelectronic applications in view of its variable band gap and lattice parameter depending upon its composition. Accurate thermophysical properties of these materials are very much needed in the semiconductor industry for the growth of large high quality crystals.
Task Progress:	The HTESL facility that was originally developed by Dr. W.K. Rhim and his coworkers at JPL was moved to the Caltech campus in October 2000. Since that time, under NASA support, the facility has undergone major upgrades both in hardware as well as in software. New software for levitation control, temperature measurements and for various thermophysical property measurements was develped using LabView. The position control software can run the control loop at the 1 KHz rate for extremely stable sample positioning. A new high power YAG laser was added for sample heating to a higher temperature. Dividing the beam into four beams of equal intensity we developed a high symmetry tetrahedral heating system which provides for uniform heating of spherical samples to elevated temperatures (2000 C and above) [3]. Thus we have achieved the capability of levitating, melting, undercooling, and processing spherical droplets of metal alloys and other materials under very near isothermal and mechanically quiescent conditions. The result is the substantial and dramatic reduction in driving forces for externally driven convective flow, gravity driven convective flow, and Marangoni convective flows within the sample by comparison with any other available platform for containerless processing of liquids [4]. The facility thus provides a unique platform for carrying out a broad range of containerless processing studies of liquids both on the ground, and potentially in microgravity. Accurate measurements of thermophysical properties of molten silicon have been attempted using an upgraded electrostatic levitator. In order to reduce the temperature gradient in the levitated sample four laser beams of equal intensity with a tetrahedral heating arrangement were used. A new image processing technique was adopted to get reliable sample volume (or density) measurements. In order to increase the accuracy of surface tension and viscosity measurements sample mass was continuously monitored throughout the experiment and the effect of mass loss was accounted for in the final analysis. Based on these new improvements, the results for the density, the specific heat over hemispherical total emissivity, the surface tension and the viscosity of molten silicon have been measured. These results were summarized and published in the Journal of Crystal Growth [2]. When containerless experiments are performed in the 1-g environment, the levitated melt is still under the influence of gravity induced effects such as convection, sedimentation and buoyancy. In addition to this, the levitation forces required in 1-g environment tend to induce oscillations. The most seriously affected by such perturbations are the transport properties such as diffusion, viscosity, surface tension, and thermal conductivity. In the case of molten germanium, whose viscosity is expected to be less than ~0.5mPa.s, our recent experiments confirmed that the levitation control force induced undesirable oscillatory instability that was almost impossible to stabilize. The germanium viscosity data shown above demonstrate this. Such oscillations not only affect the transport properties, they also seriously affect measurements of other properties such as the specific heat and the specific volume or the density. In the microgravity environment all these disturbances are expected to be substantially reduced (by orders of magnitude). If it is necessary the position control forces can be completely turned off over a short measurement period during which the sample will experience a truly quiescent condition. Molten silicon and germanium samples show such high vapor pressures that it is desirable to shorten the processing times to prevent substantial mass losses by evaporation. Short processing time is also desired if one considers the amount of coatings on the surrounding optical windows. However, it is difficult to achieve this in the ground base since rapid heating often translates to momentary loss of sample charges and subsequent loss of sample position control. In the absence of gravity the sample can be quickly heated beyond its melting temperature where strong thermionic charging mechanism provides ample charges that are required for position control. The proposed experimental facility in space has to be a form of electrostatic levitators (ESL). In ESL the electric force for sample positioning is controlled by feedback control. An important characteristic of the feedback system is that the control comes in only when it is needed, otherwise, it will virtually disappear. In the micro-g condition the control-forces will virtually disappear and produce a quiescent force environment once the sample is brought to the predetermined point in the chamber. Therefore, the Electrostatic Levitation Furnace (ELF) that is being developed by the Japan Aerospace Exploration Agency (JAXA) for the Japanese Experiment Module (JEM) in the ISS or a similar ESL furnace will be the right facility for the thermophysical property measurements in the microgravity condition.

Bibliography Type:	Description: (Last Updated: 03/30/2006)
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